



MERGING TWO SORTED ARRAYS 2

[Problem](#)[Submissions](#)[Leaderboard](#)

RATE THIS CHALLENGE



The merged array should also be sorted in non-decreasing order. Complete the function `mergeSortedArrays` that takes the following parameters:

`int arr1[n]`: the first sorted array `int arr2[m]`: the second sorted array Returns `int[]`: the merged sorted array

Input Format

The first line contains an integer n , the size of the first array. The second line contains n space-separated integers representing the first sorted array. The third line contains an integer m , the size of the second array. The fourth line contains m space-separated integers representing the second sorted array.

Constraints

$1 \leq n, m \leq 10^5$ $-10^9 \leq arr1[i], arr2[i] \leq 10^9$ Both arrays are sorted in non-decreasing order.

Output Format

Print the merged sorted array as space-separated integers.

Sample Input 0

```
4
1 3 5 7
4
2 4 6 8
```

Sample Output 0

```
1 2 3 4 5 6 7 8
```

[Change Theme](#)

Language

C



```
1  #include <stdio.h>
2  void mergeSortedArrays(int arr1[], int m, int arr2[], int n, int result[]) {
3      int i = 0, j = 0, k = 0;
4      while (i < m && j < n) {
5          if (arr1[i] <= arr2[j]) {
6              result[k++] = arr1[i++];
7          } else {
8              result[k++] = arr2[j++];
9          }
10     }
11     while (i < m) {
12         result[k++] = arr1[i++];
13     }
14     while (j < n) {
15         result[k++] = arr2[j++];
16     }
17 }
18 int main() {
19     int m, n;
20     int arr1[10000], arr2[10000];
21     int result[20000];
```

```
--
22     scanf("%d", &m);
23     for (int i = 0; i < m; i++) {
24         scanf("%d", &arr1[i]);
25     }
26     scanf("%d", &n);
27     for (int i = 0; i < n; i++) {
28         scanf("%d", &arr2[i]);
29     }
30     mergeSortedArrays(arr1, m, arr2, n, result);
31     for (int i = 0; i < m + n; i++) {
32         printf("%d ", result[i]);
33     }
34     return 0;
35 }
36
```

Line: 2 Col: 1

 Upload Code as File☐ Test against custom input

Run Code

Submit Code

[Blog](#) | [Scoring](#) | [Environment](#) | [FAQ](#) | [About Us](#) | [Helpdesk](#) | [Careers](#) | [Terms Of Service](#) | [Privacy Policy](#)